

XIANGBO ZHANG

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EDUCATION

Georgia Institute of Technology

B.S. in Mathematics

GPA: 4.00/4.00; Faculty Honors (Highest Honor for Undergrads)

Sep. 2025 – July 2026 (Expected)

Atlanta, GA

Kean University

B.A. in Mathematical Sciences & B.S. in Computer Science (Transfer Out)

GPA: 3.99/4.00; Honors Distinction in Research Track

Sep. 2023 – May 2025

Union, NJ

SELECTED COURSEWORK

Undergraduate-level: Deep Learning & Recommender System; Machine Learning; Abstract Algebra; Real Analysis; Non-Euclidean Geometry; Cryptography & Cryptanalysis; Numerical Analysis; Complex Analysis.

Graduate-level: Spectral Methods; Data Analysis & Visualization; Industrial Mathematics.

SELECTED PUBLICATIONS

Memory Dial: A Training Framework for Controllable Memorization in Language Models. **Zhang, X.**, Emami, A. *ACL ARR 2026 (January submission)*.

Stable and Explainable Personality Trait Evaluation in Large Language Models with Internal Activations. Ma, X., **Zhang, X.**, Weng, Z. *ACL ARR 2026 (January submission)*.

ES-GRPO: Evidence-Shaped GRPO for Low-Data Multi-Hop Question Answering. Hu, K., **Zhang, X.**, Tan, J. Y., Ma, X. *ACL ARR 2026 (January submission)*.

UniHash: Unifying Pointwise and Pairwise Hashing Paradigms for Seen and Unseen Category Retrieval. Ma, X., Li, R., Liu, H., **Zhang, X.**, Weng, Z. *CVPR 2026 (under review)*.

Discrete Wavelet Transform-Based Capsule Network for Hyperspectral Image Classification. Gao, Z., Wang, J., Shen, H., Dou, Z., **Zhang, X.**, Huang, K. *arXiv preprint, 2025*.

Global Output Regulation for Uncertain Feedforward Nonlinear Systems With Unknown Nonlinear Growth Rate. Chang, L., **Zhang, X.** *International Journal of Robust and Nonlinear Control (IJRNC), 2025*.

SELECTED RESEARCH EXPERIENCE

The Linear Geometry of LLM Personality

Fall 2025

Supervisor: *Prof. Zhenyu Weng*

- Developed a linear representation theory of LLM personality, modeling personality traits as low-dimensional persona vectors in transformer activation space rather than questionnaire-based self-reports.
- Justified a Local Linearity Assumption showing that personality steering operates approximately linearly in typical operating regions, enabling principled addition and subtraction of traits via vector arithmetic.
- Introduced Representation-Based Personality Measurement (RBPM) and demonstrated that stable trait subspaces emerge, with axes closely aligned to the Big Five under criteria of stability, orthogonality, and behavioral predictiveness.

Central Limit Theorem for the Binomial Distribution

Fall 2025

Supervisor: *Prof. Jeremy Zurcher*

Georgia Tech

- Conducted a theoretical study on why the Binomial distribution transitions into a Gaussian shape as sample size grows, drawing on classical asymptotic analysis.
- Used refined approximations to show combinatorial expressions simplify into a universal smooth bell curve.
- Established convergence guarantees by verifying uniform control of approximation error across all parameters.

Natural Language to Robotarium Code Generation

Supervisor: *Prof. Sam Coogan*

Fall 2025
Georgia Tech

- Built an end-to-end NL→TL→Robotarium pipeline that converts natural-language task descriptions into executable multi-robot control scripts, integrating LLMs with STL-based planning.
- Improved robustness of the code-generation workflow by refining prompt strategies and enforcing syntactic and solver-level constraints, significantly reducing failure modes in STL-to-Gurobi execution.
- Evaluated LLM-driven control generation against the GenSwarm framework, identifying architectural bottlenecks and demonstrating scenarios where direct model prompting outperformed multi-agent pipelines.

Automated Dementia Behavior Analysis and Empathetic Mapping

Supervisor: *Prof. Malihe Aliasgari*

Fall 2024–Spring 2025
Kean University

- Built a scalable multimodal analysis pipeline for dementia-related behavior research, preprocessing 100+ real-world audio–video recordings and enabling multilingual affective analysis across 90+ languages using OpenAI Whisper and Python.
- Applied large language models to extract latent emotional triggers and conversational behavior patterns, supporting automated empathetic mapping of dementia patient–caregiver interactions.
- Employed unsupervised statistical learning (K-Means, PCA) to cluster dementia conversations, systematically optimizing cluster quality via Silhouette Scores and revealing interpretable emotional–behavioral structures.

DWT-CapsNet for Hyperspectral Image Classification

Supervisor: *Prof. Zhiqiang Gao*

Summer 2024–Fall 2024
Kean University

- Developed a wavelet-driven capsule network architecture integrating attention-guided downsampling and multi-scale routing to capture fine-grained spectral–spatial representations in hyperspectral imagery.
- Designed an efficient capsule connectivity strategy that prunes redundant routes, significantly reducing FLOPs, parameters, and runtime without compromising accuracy.
- Achieved state-of-the-art performance on standard HSI benchmarks, demonstrating the model’s practical viability for large-scale earth observation and remote sensing applications.

Global Output Regulation for Uncertain Feedforward Nonlinear Systems

Supervisor: *Dr. Le Chang*

Summer 2024–Fall 2024
Shanghai University of Electric Power

- Investigated output regulation for feedforward nonlinear systems whose nonlinear terms have an entirely unknown input-dependent growth rate, significantly relaxing classical Lipschitz-type assumptions.
- Helped design time-varying and adaptive output-feedback controllers that bound the input, transform the unknown rate into an equivalent constant, and ensure global regulation via Lyapunov and dynamic-gain analysis.
- Validated the proposed control framework on nonlinear circuit and polynomial benchmark systems, demonstrating robust regulation performance under highly uncertain growth dynamics.

TEACHING EXPERIENCE

STME Tutor, Educational Opportunity Fund (EOF) Program

Supervisor: *Ms. Donna Hunter*

Fall 2024 – Spring 2025
Kean University

- Provided tutoring in computer science and math, focusing on programming, calculus, and algebra for first-generation and underrepresented students.
- Covered CS topics including class design, inheritance, polymorphism, recursion, and debugging, emphasizing rigorous problem-solving and algorithmic thinking.
- Developed practice materials and exam strategies to strengthen conceptual understanding in CS and math.

Head Teaching Assistant, MATH 1010 Foundations of Math

Supervisor: *Prof. Sangeet Srivastava*

Summer 2024
Kean University

- Led weekly studio sessions, clarified course concepts, and hold weekly Office Hours.
- Managed grading, assignment review, and feedback delivery for a class of 80+ students.
- Coordinated with the course instructor to refine teaching materials and streamline student support structure.

HONORS AND AWARDS

First Class Dean’s Scholarship (Top 3%)
Study Abroad Scholarship (Top 5%)

Zhejiang Province Government Scholarship (Top 3%)
Dean’s List (Top 15%)